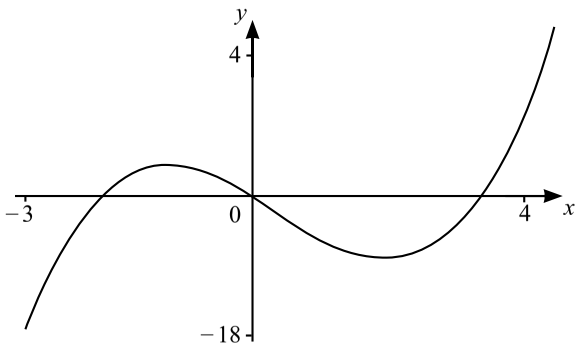
Question	Answer	Marks	Partial Marks
22	$\frac{16}{\sqrt{3}}$ or $\frac{16\sqrt{3}}{3}$	4	**B1** for $\cos 30 = \frac{\sqrt{3}}{2}$ **M2** for $\frac{8}{\cos 30}$ or **M1** for $\frac{8}{x} = \cos 30$
23	$\frac{25 - x}{(x - 4)(x + 3)}$ oe final answer	3	**M1** for denominator $(x – 4)(x + 3)$ or $x^2 – x – 12$ **B1** for $3(x + 3) – 4(x – 4)$ seen
24(a)	$(x – 2)^2 + 3$	2	**M1** for $(x – 2)^2$ seen
24(b)	(2, 3)	1	**FT** *their* **(a)**
25	$\frac{15}{2}$ oe	3	**M2** for $\frac{9}{x} = \frac{6}{5}$ oe or $x^2 = \frac{25}{36} \times 9^2$ oe or better or **M1** for $\sqrt{\frac{36}{25}}$ or $\sqrt{\frac{25}{36}}$ oe or better or for $\left(\frac{9}{x}\right)^2 = \frac{36}{25}$ or $\left(\frac{x}{9}\right)^2 = \frac{25}{36}$ oe or better
26(a)	Correct sketch and –2, 0 and 3 indicated on x-axis with graph going beyond intersections with x-axis  	3	**B2** for correct sketch without labels or **B1** for any positive cubic with two turning points
26(b)	$x^3 – x^2 – 6x$	3	**B2** for $x(x^2 + 2x – 3x – 6)$ or $(x^2 + 2x)(x – 3)$ or $(x + 2)(x^2 – 3x)$ or **B1** for $x^2 + 2x – 3x – 6$ or $x^2 + 2x$ or $x^2 – 3x$

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Question	Answer	Marks	Partial Marks
26(c)	(0, –1)	**5**	**B2** for $3x^2 - 2x - 6$ or **B1** for $3x^2$ or $2x$ or – 6 **B2** for $y = -5x - 1$ or **M1** for substituting 1 into *their* derivative and **M1dep** for $-6 = (\textit{their}\ -5) \times 1 + c$ or use of rise/run to find rise **FT** *their* **(b)** for the whole of **(c)**

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